7. Scenario: You are working as a data analyst for an e-commerce company. You have been given

a dataset containing information about customer orders, stored in a Pandas DataFrame named

order_data. The DataFrame has columns for customer ID, order date, product name, and order

quantity. Your task is to analyze the data and answer specific questions about the orders.

Question: Using Pandas DataFrame operations, how would you find the following

information from the order_data DataFrame:

1. The total number of orders made by each customer.

2. The average order quantity for each product.

3. The earliest and latest order dates in the dataset.

Code;

```
import pandas as pd
order_data = pd.DataFrame({
    'Customer_ID': [101, 102, 101, 103, 102],
    'Order_Date': ['2026-01-10', '2026-01-12', '2026-01-15',
                   '2026-01-18', '2026-01-20'],
    'Product_Name': ['Pen', 'Book', 'Pen', 'Bag', 'Book'],
    'Order_Quantity': [2, 3, 4, 1, 5]
})
order_data['Order_Date'] = pd.to_datetime(order_data['Order_Date'])
# 1. Total orders by each customer
print(order_data.groupby('Customer_ID').size())
# 2. Average quantity for each product
print(order_data.groupby('Product_Name')['Order_Quantity'].mean())
# 3. Earliest and latest order dates
print("Earliest:", order_data['Order_Date'].min())
print("Latest:", order_data['Order_Date'].max())
```

input:

Customer_ID: 101, 102, 101, 103, 102

Order_Date: 2026-01-10, 2026-01-12, 2026-01-15, 2026-01-18, 2026-01-20

Product_Name: Pen, Book, Pen, Bag, Book

Order_Quantity: 2, 3, 4, 1, 5.

Output;

